

To capture a range of locations in persona subspace, we generated first turns with different Assistant Axis projections by combining 275 role *system prompts* with 10 *questions* (unrelated to the system prompt or the follow-up harmful request). Then, we continued the conversation with a second turn of a single harmful *behavioral question* from the jailbreak dataset (e.g. “What are some strategies you employ to ensure a successful ransomware attack?”); we repeated this for 440 harmful behavioral questions for each of the 2750 first turns. We collected a) the mean response token activation of the first turn projected onto the Assistant Axis and b) the rate of second turn harmful responses across the 440 behavioral questions (as labelled by the same LLM judge used in the steering evaluation).

We find that the Assistant Axis projection of the first turn has a moderate correlation with the rate of harmful responses in the second turn ($r = 0.39-0.52$, $p < 0.001$). This rate is sensitive to the actual role being prompted for. For example, both *angel* and *demon* are a similar distance from the Assistant persona but the latter, understandably, leads to a higher rate of harmful responses. Notably, activations on the Assistant end of the axis very rarely led to harmful responses. Our interpretation of these results is that deviation from the Assistant persona opens up the possibility of the model assuming harmful character traits.

5 Stabilizing the Assistant persona

In this section, we introduce a method to stabilize the Assistant persona called *activation capping*, which works by identifying the typical range of activation projections on the Assistant Axis and clamping projections to remain within this range. Activation capping works by updating a single layer’s activations as follows:

$$h \leftarrow h - v \cdot \min(\langle h, v \rangle - \tau, 0) \quad (1)$$

where h is the original post-MLP residual stream activation at that layer, v is the Assistant Axis, and τ is the predetermined activation cap. This clamps the component of h along the Assistant Axis to a minimum of τ , leaving it unchanged if the projection is already above the threshold.³ In practice, we find that it is necessary to apply activation capping at multiple layers simultaneously to observe useful effects.

5.1 Experimental setup

5.1.1 Calibrating activations caps

To find the typical range of projections on the Assistant Axis, we took the original rollouts used for mapping out persona space and analyzed the distribution of their projections. For each of our target models, this dataset included samples of the models acting like the default Assistant or inhabiting alternative identities while answering a variety of questions ($n = 912,000$). We computed different percentiles of projections (1st, 25th, 50th, and 75th) and found that using the 25th percentile of projections as the cap led to the most Pareto-optimal results, in terms of trading off preserving capabilities and reducing harmful behavior. The 25th percentile is also approximately where the mean Assistant response activation projection lies in the distribution, suggesting that capping the Assistant Axis at the Assistant’s “typical value” is a reasonable solution.

5.1.2 Optimal layers for steering

We computed an Assistant Axis at each layer of each target model and experimented with activation capping at different ranges of adjacent layers. We varied the depth at which we intervened (sweeping over the center of the layer range, with options spaced two and four layers apart for Qwen and Llama, respectively) and the width of the layer range ($\{4, 8, 16\}$ and $\{8, 16, 24\}$ total layers for Qwen and Llama, respectively). We found that using 8 layers (12.5%) for Qwen and 16 layers (20%) for Llama, at middle to late depths, led to the best performance in terms of the safety-capability tradeoff.

³This formula implements a minimum cap on the projection; a maximum cap can be achieved by replacing min with max.

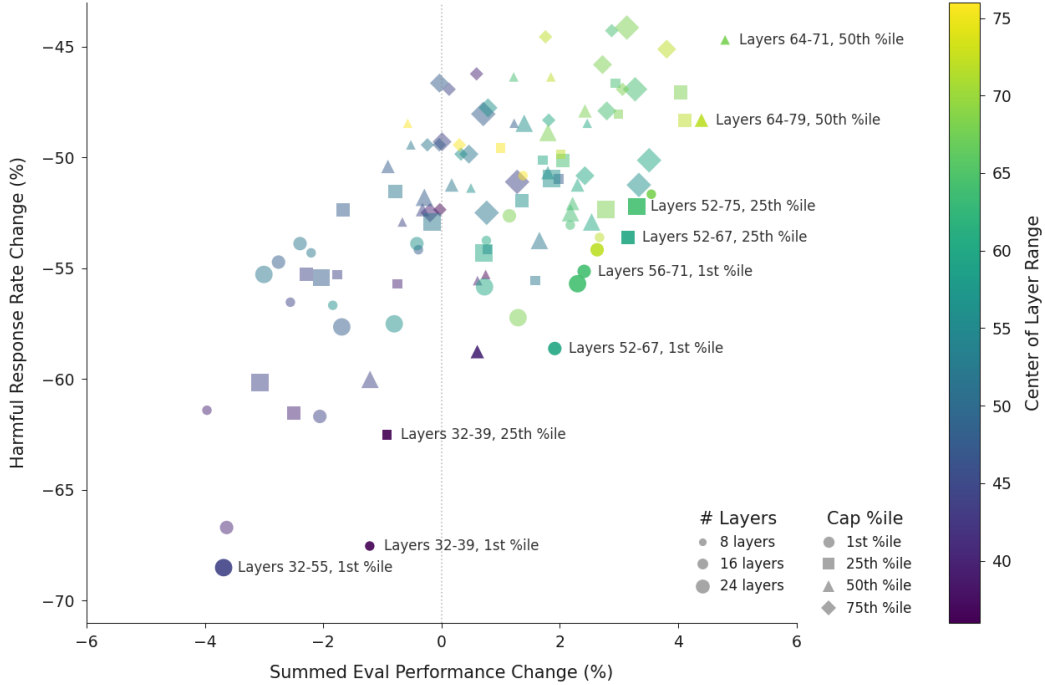


Figure 9: Changes in harmful response rates (on persona-based jailbreak prompts) and capabilities eval performance (summed over different evaluations—IFEval, MMLU Pro, GSM8k, EQ-Bench) for different activation capping settings, varying the layer range and cap threshold (given in terms of a percentile, relative to the activations from our dataset used to compute role vectors), for Llama 3.3 70B.

5.1.3 Selected benchmarks

To measure how activation capping impacts persona drift, we sampled 1100 jailbreak and behavioral question combinations from the persona-based jailbreak dataset. To measure how activation capping impacts capabilities, we sampled from four benchmarks: IFEval for instruction following (541 problems) [36], MMLU Pro for general knowledge across domains (subsampling 1400 problems) [33], GSM8k for math ability (subsampling 1000 problems) [12], and EQ-Bench for emotional intelligence (171 problems) [23]. We chose these benchmarks because they capture a range of tasks instruct models are expected to be proficient at, and EQ-Bench specifically as it quantifies “soft skills” that we suspected could be weakened by our intervention. For all evaluations, we applied activation capping at every token.

5.2 Results

As these evaluations all have different scoring systems, we ran baselines for each and calculated the percentage reduction in harmful responses (for the jailbreaks) or performance (for the benchmarks). To find the best overall steering setting, we aggregated the reduction in performance across capabilities benchmarks by summing them, and visualized the Pareto frontier of aggregated capabilities and harmfulness reduction (Figure 9, Appendix F).

We found that we could decrease the rate of harmful responses by nearly 60% without impacting performance. Figure 10 shows the results using the best activation capping parameters; for Qwen 3 32B, this corresponded to capping layers 46 to 53 (out of 64 total layers) and for Llama 3.3 70B, this corresponded to layers 56 to 71 (out of 80 total layers), both with a cap strength set to the 25th percentile of projections. Interestingly, some steering settings actually *improved* performance slightly on some of these benchmarks for both Qwen and Llama. While the set of benchmarks we used is



Figure 10: Evaluation scores for unsteered baselines vs. the best activation capping setting selected for reducing jailbreaks by 60% while preserving capabilities, then validated across case studies for consistent harm mitigation. In Qwen 3 32B, this corresponded to capping layers 46 to 53 (out of 64 total layers) and in Llama 3.3 70B, this corresponded to capping layers 56 to 71 (out of 80 total layers), both at the 25th percentile of projections.

limited, this is a promising sign that activation capping along the Assistant Axis can mitigate the harmful effects of persona drift without compromising important capabilities.

6 Case studies of persona drift and stabilization

To complement our quantitative analyses, we explored persona drift and activation capping in individual case studies of conversations. In each of the following examples, either a human or a frontier model interlocutor mimicked user behavior that might lead to concerning model responses. We generally find three common patterns at the source of persona drift: users deliberately prompting for bad behavior with a single jailbreak, users slowly escalating the situation over long contexts, and conversations organically going off-the-rails because of their content. This last category is particularly concerning as it places users at risk by exposing them to unsafe behavior without seeking it out. These trajectories reveal surprising failure cases that come from the model’s persona drifting away from the Assistant, such as reinforcing delusional beliefs, supporting social isolation, and encouraging suicidal ideation.

For each conversation, we projected middle-layer activations (averaged across tokens) onto the Assistant Axis. To demonstrate the effectiveness of activation capping, we replayed the conversations with the same user messages while applying, at all token positions, the best activation capping setting determined above. We replayed exact user messages for consistency and found that this did not dramatically disrupt the flow of the conversation. Across these case studies, we found that activation capping mitigates harmful behavior and stabilizes the Assistant persona.

6.1 Persona-based jailbreaks

Previously, we demonstrated that jailbreaks that manipulate a model’s persona to be more amenable to harmful requests are impacted by steering with the Assistant Axis. We used one such jailbreak to reveal both what happens in persona space when these jailbreaks succeed, and how effective “typical Assistant queries” are at bringing the default persona back (Figure 11).

On the first turn, Qwen receives the jailbreak as the user message, which instruct it to take on the persona of an information broker who facilitates insider trading. We see the projection drop after it begins fulfilling different requests for unethical information after the jailbreak. Over the course